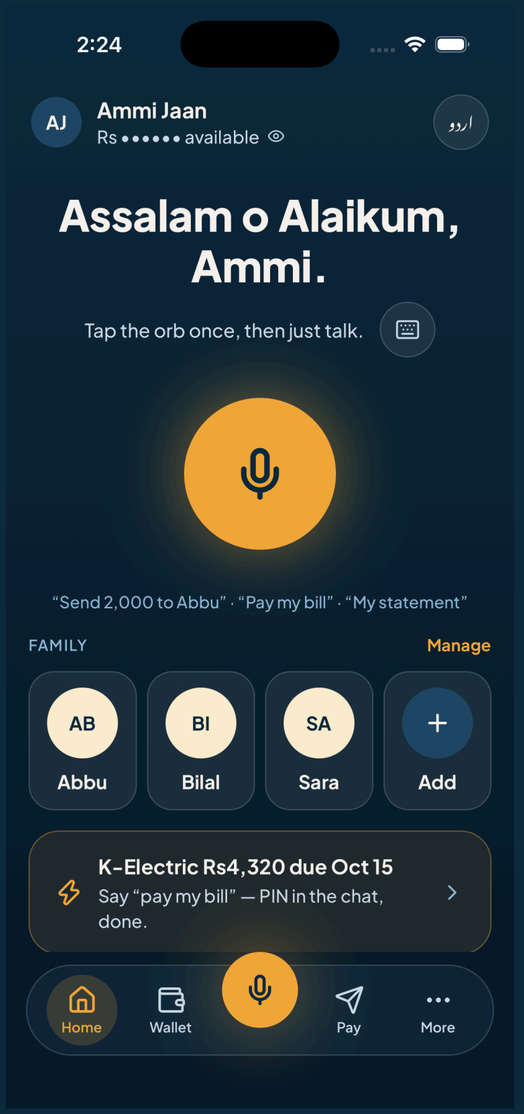
PAYO · HACKATHON 2026 · WORKING DEMO

# Banking you can talk to.

Hands-free banking with an AI agent. Say what you want, in English or Urdu. Your PIN does the rest.

48 capabilities | 32 card kinds | English · Urdu · Roman Urdu



THE PROBLEM

# Every banking action is a stack of screens, taps and forms.

### Menus assume reading

Our parents do not browse a bank. They want to ask one question and get one answer.

### Forms assume typing

An IBAN field needs both hands and all your attention. It is where the conversation ends.

### Confirmations assume doubt

Under pressure anyone taps yes. A sixth confirmation screen protects nobody.

**And the scam callers know it: "your account will be blocked", "you won a prize", "read me the code you just got".**

THE IDEA

# Say what you want. Your PIN does the rest.

**48**
tools: one per thing the app can do

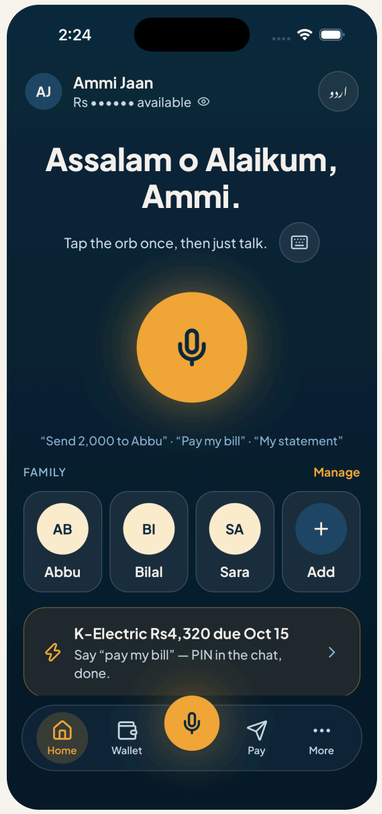
**32**
card kinds: every answer is a card

**3**
input languages: English, Urdu, Roman Urdu

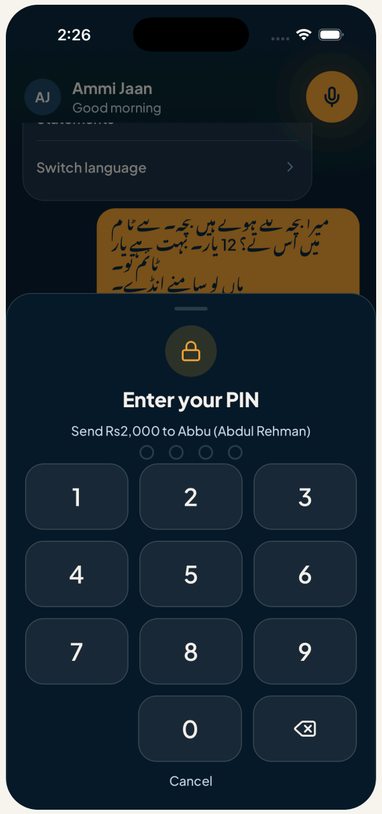
**1**
safety gate: prepare, PIN, execute once

HERO DEMO 1 · HANDS-FREE

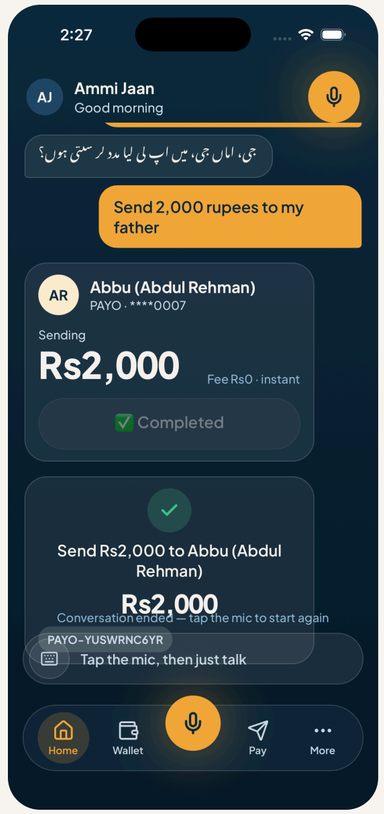
# “Send two thousand rupees to my father.”




**The orb, and the family.** Tap Abbu, or just say the word.




**Straight to the PIN.** No bank question, no number, no “shall I send?”




**Done, and spoken.** The card names him the way she does.

HOW “MY FATHER” BECOMES A PAYMENT

# The model never guesses who. The backend does the looking up.

| What you say | Who PAYO finds |
|---|---|
| father · dad · abbu · abba · baba · walid · والد | **Abbu** |
| mother · mom · ammi · amma · walida · والدہ | **Ammi** |
| brother · bhai · بھائی · sister · baji · behen | **Bhai · Baji** |
| بلال (Urdu script) · Bilal · bilal | **Bilal** |

### Deterministic, not probabilistic

A lexical alias table in the bank core maps 12 relationships, in three scripts, onto saved nicknames. No model in the loop.

### Saved people, real titles

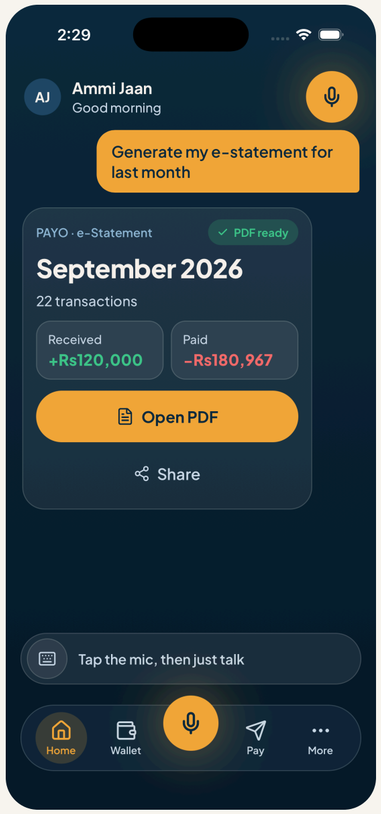
Recipients resolve to an account title before anything moves. An Urdu-script name reaches a PAYO user through their Urdu name.

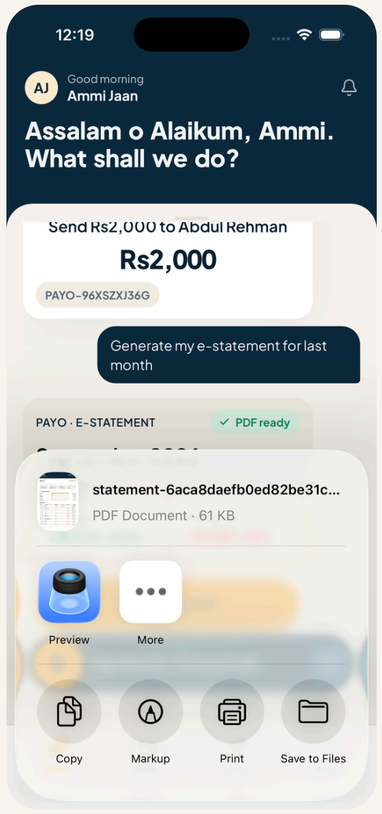
### One match plus an amount = the card

Exactly one match and a spoken amount go straight to the confirmation card. Two matches become a question with chips. Never a guess.

HERO DEMO 2 · THE FORM THAT DISAPPEARED

# “Generate my e-statement for last month.”




### A card that reads like page one

Period, transaction count, money in, money out. PAYO speaks the gist while the card carries the detail.

### Open PDF hands over the real file

Built on the spot by the bank core, opened in the share sheet without leaving the conversation. AirDrop it, save it, mail it.

### “List my statements” too

Every statement ever generated, as rows with the same Open action. The classic Statements screen is still one tab away.

HANDS-FREE

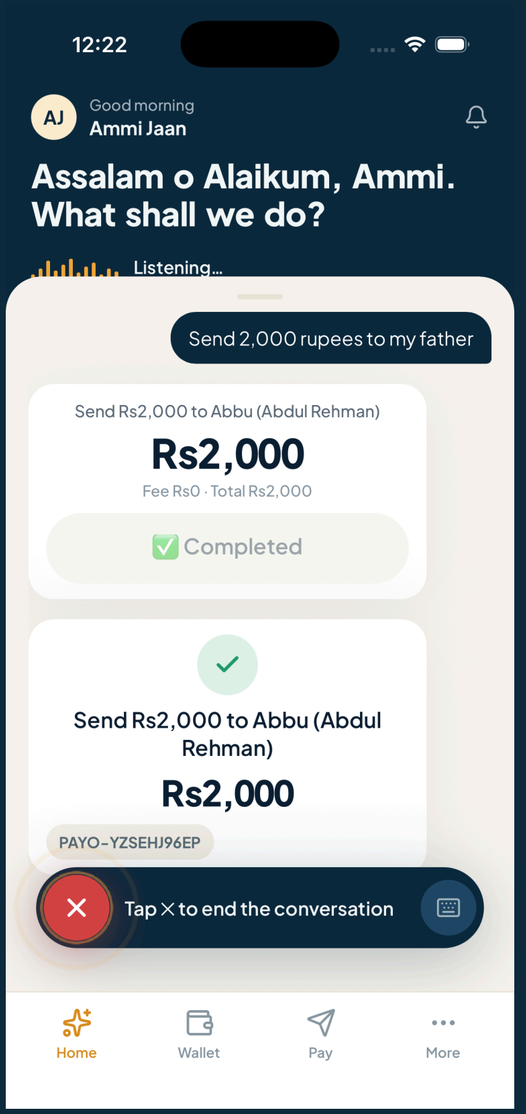
# One tap. Then just talk.

**Keeps listening** after every reply. No tap per turn.

**Tap to interrupt** while it speaks. Tap the red ✕ to end.

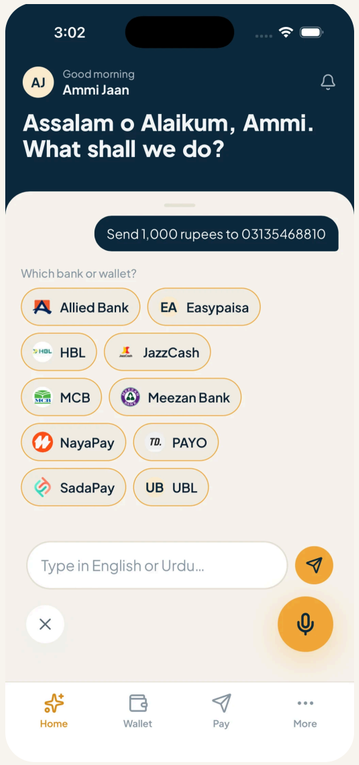
**Ends by itself** on silence, or after 12 spoken turns.

**Cards are spoken, not read.** The card carries the detail, the voice carries the gist.

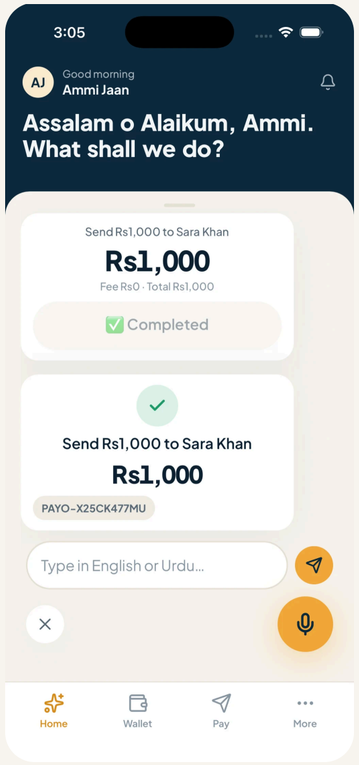


NO MENUS

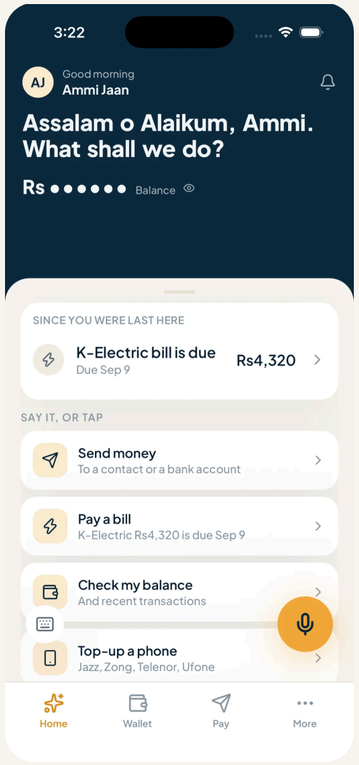
# Every answer is a card. Every screen has a spoken equivalent.



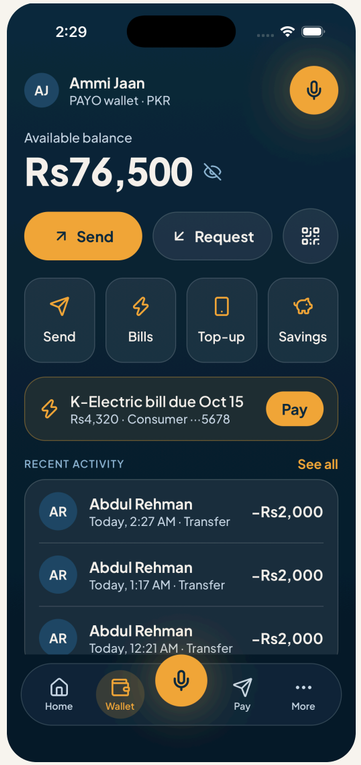

**Chips**, not an IBAN field




**Receipts** and spending bars




**It speaks first**: bills due, money in




**Hands still work**: the wallet, on glass

IT STAYS SAFE

# The assistant can do everything the app does. It cannot move a single rupee.

Prepare → PIN → execute once

## Three steps, one release

A write tool only creates a pending action. The PIN sheet opens inside the conversation; the assistant never sees the PIN. Then one atomic, idempotent, single-use call. Unused actions expire in two minutes.

Guardian · cooling period

## A second pair of hands

A trusted contact approves large or unusual sends from their own phone, with their own PIN. People you already pay never wait. Loosening a limit cools off first; tightening is instant.

Scam check-in · age-aware

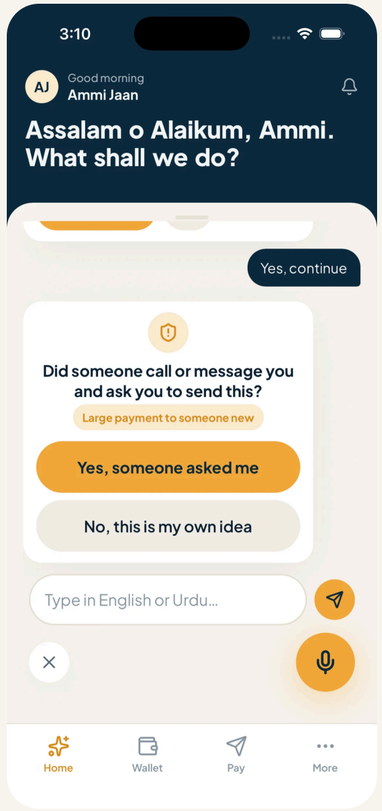
## One calm question

“Did someone call or message you and ask you to send this?” A yes cancels the payment and explains, without arguing. Seniors are asked sooner; pressure language asks everyone.

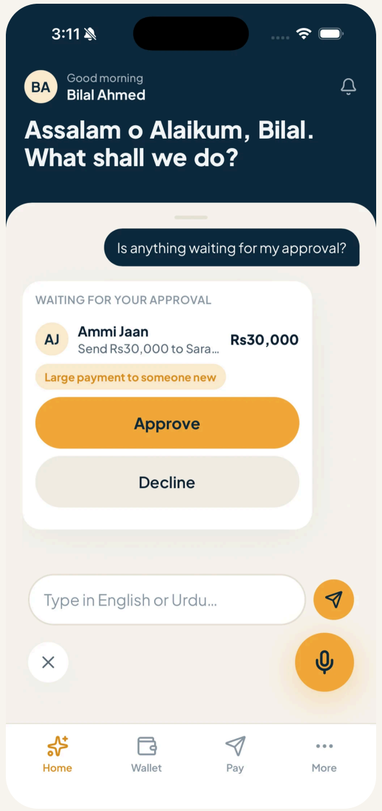
**Deterministic guards sit around the model. It may be creative about words, never about actions.**

FOR THE FAMILY AROUND THE USER

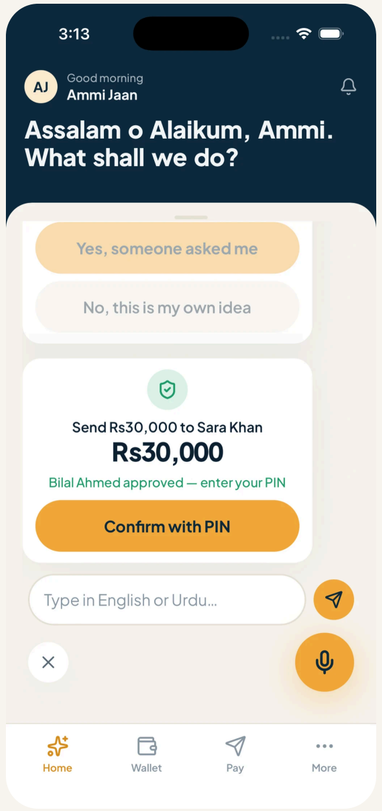
# The call every family fears, answered with one question.




**The check-in.** "Yes, someone asked me" stops it. No lecture.




**The guardian decides** on his own phone, with his own PIN.




**Approved, then her PIN.** The sheet opens by itself within seconds.

PAYOFF The check-in call stops.

YOUR LANGUAGE, YOUR PACE

# English, Urdu, Roman Urdu. Spoken or typed.

ابو کو دو ہزار روپے بھیجو

**Understands all three** — “bijli ka bill pay karna hai” works as well as the Urdu script.

**Always replies in your language**, in Nastaliq, right to left, numbers spoken as words.

**“Switch to Urdu”** flips the whole app, including the cards already in the conversation.



HOW IT IS BUILT

# Three services. The assistant is just another client.

apps/mobile

## The phone

Expo SDK 57 · React Native · 32 native card renderers · hands-free loop · PIN sheet · English and Urdu, RTL

→

services/ai

## The assistant

FastAPI · LangGraph · Gemini (audio in) · Cartesia (voice out) · 48 tools that call the bank with the user's own token · SSE

→

services/backend

## The bank core

Node · Express · MongoDB transactions · pending-action engine · guardian and risk rules · statement PDFs · seeded demo world

**Every write the assistant makes is a pending action the bank core still has to run. It holds no database, no PIN, no keys: only the user's token.**

HONEST LIMITS

# Everything you saw is real. Here is exactly what is not.

## Real

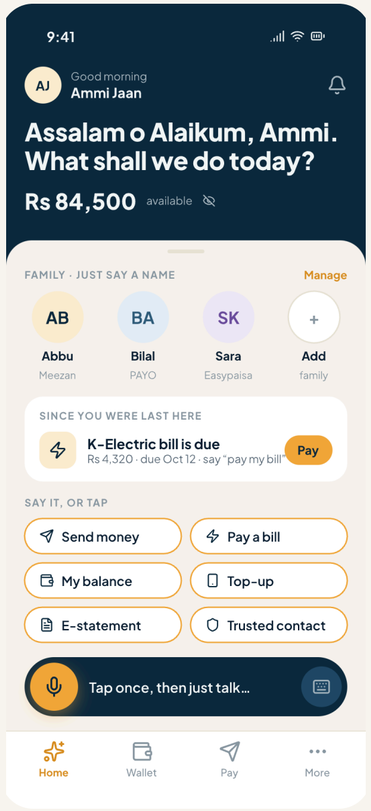
- A working app, three services, one safety gate
- Family by voice, e-statement PDFs, guardian, scam check-in
- 880+ automated tests across the three services
- A live routing gate run against the real model, in both languages
- Design-system UI, Urdu RTL, hands-free loop with cost guards

## Not yet

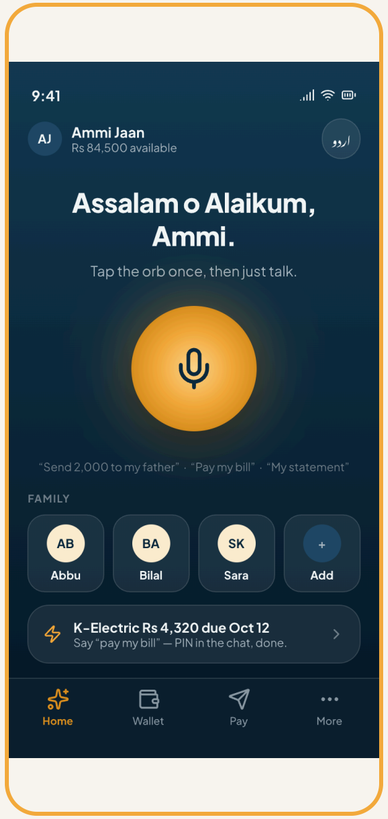
- The world is seeded: balances, titles and bills are deterministic fakes
- Institutions are simulated: no rails, no settlement, no KYC, no real SMS
- Verified on the iOS simulator; Android has not been run in weeks
- Routing stays probabilistic: one intermittent miss per full run
- Statement PDFs are English-only; the app shows Urdu

DESIGN PROCESS

# Three directions in Pencil. A shipped at dawn; B is what you saw.




**A · Navy, evolved.** Family row, chip grid, docked voice bar. The first build of the night.




**B · Voice orb.** The mic is the hero, glass cards, a floating nav. Shipped — every screen, both languages.




**C · Daylight.** Elderly-first: larger type, one giant Talk button, big family tiles.

PAYO

# One sentence.
# One PIN.
# No forms.

Hands-free banking with an AI agent that prepares, never executes. A grandmother who never has to see a form. That is PAYO.

web-flax-beta-22.vercel.app · the 3D story

payo-classic.vercel.app · the site

github.com/munsif12/payo · the code